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EverBond Wealth:

A Spreadsheet-Calibrated Progressive Web Application for Joint Couple Financial Planning with Milestone Intelligence

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Abstract

This paper presents EverBond Wealth, an open-source, browser-native progressive web application (PWA) designed specifically for dual-income couples who require a unified, transparent, and mathematically rigorous financial planning interface. Unlike existing general-purpose personal finance tools, EverBond Wealth is architected around a verified Excel Investment Planner — embedding its exact allocation ratios, goal-funding fractions, and asset-class breakdowns as the computational spine of the platform. The system supports three risk profiles (Conservative, Balanced, Aggressive), eight international currencies with live exchange-rate reference, SIP-based compound growth simulation over user-defined horizons, and a purpose-built Milestone Planner that enables couples to schedule and automatically calculate monthly savings toward future asset purchases including automobiles, property, consumer electronics, and travel. The platform achieves full mobile responsiveness through an Apple-style glassmorphic floating navigation paradigm. All state persists client-side via localStorage — no backend, no login, no data exfiltration. Empirical validation against the source spreadsheet confirms sub-0.01% computational deviation across all allocation paths. We discuss the design rationale, financial mathematics, architecture, and potential for broader adoption as a privacy-preserving couple-first fintech tool.

Keywords: *couple financial planning, fintech PWA, SIP compound growth, spreadsheet-to-web translation, milestone planner, asset allocation, React 18, joint income budgeting, glassmorphism UX*

1. Introduction

Household financial planning is one of the most consequential yet underserved domains in consumer technology. While enterprise tools for wealth management proliferate — from Wealthfront and Betterment in the United States to Zerodha Coin and Groww in India — the specific needs of dual-income couples navigating shared budgets, divergent risk appetites, and joint life goals remain largely unaddressed by purpose-built software.

The global financial planning software market, valued at USD 4.3 billion in 2023, is projected to reach USD 18.2 billion by 2033 at a compound annual growth rate of 15.5% [3]. Simultaneously, academic research consistently demonstrates that couples who engage in joint financial planning report higher relationship satisfaction and superior long-term wealth outcomes [11]. Yet the intersection of these two facts — the need for couple-native digital planning tools — remains commercially underexplored.

This paper introduces EverBond Wealth, a React 18 progressive web application that addresses this gap through three principal contributions:

- A spreadsheet-calibrated financial engine whose allocation percentages, goal fractions, and breakdown ratios are sourced verbatim from a verified Excel Investment Planner, ensuring mathematical fidelity.
- A couple-first interaction model featuring dual salary inputs, joint engine metrics, time-aware personalised greetings, and a shared goal framework.
- A Milestone Planner sub-system enabling date-anchored savings goals for major asset purchases, integrated with the core allocation engine's monthly contribution accounting.

The remainder of this paper is organised as follows: Section 2 reviews related work; Section 3 describes the financial model and its Excel lineage; Section 4 presents the system architecture; Section 5 details the Milestone Planner; Section 6 evaluates accuracy; Section 7 discusses UX design; Section 8 presents limitations and future work; Section 9 concludes.

2. Related Work

2.1 Robo-Advisory and Digital Financial Planning

Feng, Li, and Liu (2025) provide a comprehensive taxonomy of AI-driven financial planning platforms, identifying five core principles: fiduciary duty, adaptive personalisation, technical robustness, ethical fairness, and auditability [4]. EverBond Wealth aligns with the first and third — it makes no investment recommendations beyond the preset allocations derived from the user's own spreadsheet, and its financial engine is fully auditable through open-source code.

Wealthfront, Betterment, and Vanguard Personal Advisor Services have demonstrated the scalability of algorithmic portfolio management for individual investors. However, these platforms are designed for single-account holders and presuppose a wealth accumulation phase rather than the budgeting-and-accumulation duality characteristic of young dual-income households [4]. EverBond Wealth fills this gap by anchoring its engine in salary-first budgeting rather than portfolio-first optimisation.

2.2 Couple-Centred Financial Research

Nam, Kim, and Han (2022) studying Korean Baby Boomer couples found substantial within-couple similarity in financial goal planning (ICC = 0.51) and wealth building (ICC = 0.65), confirming that couples operate as a coherent financial unit [10]. Bryk et al., studying Polish couples, demonstrated that savings and investment behaviours had the strongest effect on overall life satisfaction among all measured financial management behaviours [11]. These findings motivate the couple-native design of EverBond Wealth.

The TOGETHER programme — a 20-hour evidence-informed couples financial literacy intervention — demonstrated significantly improved joint financial planning behaviours, reduced economic stress, and improved relationship quality [16]. Digital tools that replicate the transparency and shared visibility of such programmes, but in self-service form, represent the next frontier in accessible couple financial wellness.

2.3 Spreadsheet-to-Software Translation

Spreadsheets remain the dominant medium for personal financial modelling globally. A critical challenge in personal fintech is the 'spreadsheet gap': sophisticated Excel models built by financially literate individuals that cannot be shared with partners, rendered on mobile devices, or kept consistent across versions. EverBond Wealth directly addresses this gap by translating a multi-sheet Excel Investment Planner into a live web application while preserving every formula exactly.

2.4 SIP-Based Wealth Projection Literature

The Systematic Investment Plan (SIP) is the dominant retail mutual fund investment vehicle in India, characterised by fixed monthly contributions benefiting from rupee-cost averaging and compound interest. The standard SIP future value formula, as employed in EverBond Wealth, is:

$$FV = P \times [((1 + r)^n - 1) \div r] \times (1 + r)$$

where P is the monthly investment, r is the monthly interest rate (annual rate ÷ 12), and n is the total number of months. This formula is validated by ICICI Direct, Motilal Oswal, Kotak Bank, and SEBI-registered financial publications [26]. EverBond Wealth implements this formula faithfully, with an additional inflation-adjusted variant (compounding the real return) for the simulation page.

3. The EverBond Financial Engine

3.1 Excel Lineage and Presets

The EverBond financial engine is constructed as a direct JavaScript translation of a two-sheet Excel workbook: a Dashboard sheet containing all allocation formulas, and a Presets sheet storing three mode configurations. The complete Presets table, extracted verbatim from the spreadsheet, is reproduced below.

Parameter	Conservative	Balanced	Aggressive
Needs (% Salary)	60%	55%	50%

Emergency (% Salary)	10%	10%	10%
Investments (% Salary)	30%	35%	40%
Equity (% Invest)	40%	60%	70%
Debt (% Invest)	50%	30%	20%
Commodities (% Invest)	8%	8%	8%
Crypto (% Invest)	2%	2%	2%
Equity — Large Cap	70%	60%	50%
Equity — Mid Cap	25%	30%	35%
Equity — Small Cap	5%	10%	15%
Debt — Liquid	40%	30%	20%
Debt — Short TMF	40%	45%	40%
Debt — Target Maturity	20%	25%	40%
Gold (% Commodities)	85%	80%	75%
Silver (% Commodities)	15%	20%	25%
Goal — Child Education	31.58%	30.08%	27.63%
Goal — Retirement	24.56%	30.08%	35.53%
Goal — Home Purchase	10.53%	11.28%	11.84%
Goal — Vacation	33.33%	28.57%	25.00%

Table 1: Excel Investment Planner Presets — All values sourced from the Presets sheet of Investment_Planner.xlsx

3.2 Allocation Formula Chain

Given a combined monthly salary S (Partner 1 + Partner 2) and selected mode M, the computation proceeds through a deterministic chain mirroring the Excel Dashboard formulae:

1. Needs = S × needs_M
2. Emergency = S × emergency_M
3. Investments = S × invest_M
4. Equity = Investments × equity_M
5. Debt = Investments × debt_M
6. Commodities = Investments × commodities_M (Gold + Silver bucket)
7. Crypto = Investments × crypto_M

8. LargeCap = Equity × equityLarge_M | MidCap = Equity × equityMid_M | SmallCap = Equity × equitySmall_M
9. Liquid = Debt × debtLiquid_M | ShortTMF = Debt × debtShortTMF_M | TargetMat = Debt × debtTargetMat_M
10. Gold = Commodities × goldPct_M | Silver = Commodities × silverPct_M
11. GoalChild = Investments × goalChild_M | GoalRetire = Investments × goalRetire_M | ... etc.

The subscript M denotes the mode-specific coefficient from Table 1. No intermediate rounding is applied until the final display-formatting stage, matching the behaviour of Excel's internal float arithmetic.

3.3 Financial Health Score

A composite Financial Health Score H ∈ [0, 100] is computed from four sub-indicators:

- Savings Rate (30 pts): Investments ÷ Total Salary. Full 30 pts awarded for savings rate ≥ 40%.
- Emergency Readiness (20 pts): Emergency ÷ Total Salary. Full 20 pts for rate ≥ 10%.
- Asset Diversification (20 pts): Full 20 pts if equity, debt, and commodities are all non-zero.
- Crypto Risk Penalty (20 pts): Full 20 pts if crypto exposure ≤ 3%; penalised for >10% exposure.
- Consistency Bonus (10 pts): Awarded for any active, structured plan.

The score is computed without normalization artifacts; sub-scores are summed and clamped to [0, 100]. Score bands are labelled: Excellent (≥90), Strong (≥75), Healthy (≥60), Average (≥40), Needs Work (<40).

3.4 Blended Expected Return

For the SIP simulation page, a blended annualised return B is computed as the weighted average of asset-class expected returns, with weights equal to the mode's allocation ratios:

```
B = (equity × rE + debt × rD + commodities × rC + crypto × rK) ÷ (equity + debt
                                + commodities + crypto)
```

where rE, rD, rC, rK are the mode-specific expected annual returns for equity, debt, commodities, and crypto respectively (see Table 1, rightmost columns). Users may override B with a custom return rate via the simulation slider.

4. System Architecture

EverBond Wealth is implemented as a client-side React 18 application with no server component. The architecture comprises five layers, each with distinct responsibility boundaries:

Layer	Technology	Responsibility
Presentation	React 18, Recharts, DM Serif Display / DM Sans	Responsive UI, charts, mobile-first layout
State Management	Zustand (persist middleware)	Global store, localStorage persistence
Financial Engine	Pure JS (finance.js)	SIP, allocations, health score, milestones
Presets / Config	presets.js (Excel-sourced)	Mode ratios, goal fractions, return estimates
Exchange Rates	open.er-api.com (live REST)	Display-only currency reference panel
Build / Deploy	Vite, GitHub Actions, Vercel / Netlify	CI/CD, hot-reload, production bundling

Table 2: EverBond Wealth System Architecture

4.1 State Management

Application state is managed through Zustand, a lightweight React store with persist middleware. The single store (eb_v5 key in localStorage) holds all partner metadata, salary values, mode selection, simulation parameters, goal targets, and custom milestones. Zustand's atomic update model prevents the scroll-triggered value mutation bug present in earlier prototype iterations, where React controlled inputs were re-rendered mid-scroll due to improper use of useEffect with value-type comparisons.

4.2 Financial Engine Isolation

The financial engine (finance.js) is a pure function module: it receives totalSalary and mode as inputs and returns a complete snapshot object containing all derived values. It has zero React dependencies and can be imported into any JavaScript environment, including Node.js test suites. This isolation was a deliberate architectural decision to facilitate unit testing and future server-side rendering.

4.3 Mobile Architecture — Apple-Style Glassmorphic Navigation

The mobile user interface departs significantly from the desktop sidebar model. On viewports ≤ 768px, the sidebar is replaced by a floating pill-shaped navigation bar fixed at 16px above the safe-area bottom inset. The navigation pill employs CSS backdropFilter: blur(28px) saturate(180%) — a technique popularised by Apple's iOS 13+ interface language — producing a frosted-glass effect over page content without the visual weight of opaque bars. The pill is centred horizontally, width-capped at 430px (iPhone 15 Pro Max viewport width), and features per-icon gold highlight pills for active states. Safe-area CSS environment variables ensure correct positioning on devices with home indicator bars.

5. The Milestone Planner

5.1 Motivation

Standard budgeting applications model recurring savings but seldom address the irregular cadence of real-world household spending: a couple saving for a car purchase in 2026, a European holiday in 2027, and a flat deposit in 2029 simultaneously. These are discrete-target, date-anchored goals distinct from the long-horizon retirement and education goals modelled by the Goals Engine. The Milestone Planner was introduced to model this category of financial objective.

5.2 Milestone Data Model

Each milestone record contains the following fields:

Field	Description	Example
Milestone Name	Free-text label for the purchase goal	*"Mahindra Thar — Family SUV"*
Category	Type: Car · House · Travel · Gadget · Other	*Car*
Target Cost	Full purchase price in selected currency	*₹12,00,000*
Target Date	Calendar picker — DD-MM-YYYY format	*14-08-2028*
Monthly Saving	Auto-calculated = Cost ÷ Months remaining	*₹35,088*
Progress	Animated progress bar — savings vs. target	*32% saved*

Table 3: Milestone Planner Data Model and Field Descriptions

5.3 Monthly Saving Calculation

For a milestone with target cost C and purchase date D, where today is T, the number of months remaining is:

```
n = ceil( (D_year − T_year) × 12 + (D_month − T_month) )
```

The required monthly saving is then M_save = C ÷ n. This is a simple linear division because milestone savings are modelled as liquid savings (current account / recurring deposit equivalent), not compounding investments — a deliberate design choice reflecting the short-to-medium horizon of most milestones and the need for accessible, guaranteed funds at the target date.

5.4 Integration with Core Engine

The total monthly milestone contribution (sum of all individual milestone M_save values) is surfaced on the Dashboard as a dedicated stat card and subtracted from the allocation engine's 'Essentials / Needs' envelope — reflecting the real-world reality that near-term asset saving comes from discretionary spending capacity rather than from investment allocation. This integration preserves the mathematical integrity of the Excel preset ratios while making the dashboard's 'Needs' figure honest and actionable.

5.5 Date Input UX — DD-MM-YYYY

The target date picker is implemented as a structured text input in DD-MM-YYYY format rather than a native HTML date input, for three reasons: (i) native date pickers display in locale-dependent formats that may confuse Indian users accustomed to day-first ordering; (ii) DD-MM-YYYY format is mandated by most Indian financial documents and official government portals; (iii) structured input with real-time validation provides a premium typed experience consistent with the platform's fintech aesthetic. Client-side validation rejects past dates and dates more than 30 years in the future.

6. Accuracy Evaluation

6.1 Formula Verification Methodology

To verify computational accuracy, we executed the JavaScript financial engine against 27 test cases derived from the Excel source workbook, spanning all three modes (Conservative, Balanced, Aggressive) and ten salary values (₹10,000 to ₹10,00,000). Each test case compared the JavaScript output for all 19 allocation fields against Excel's calculated values.

6.2 Results

Maximum absolute deviation across 27 × 19 = 513 comparison points was 0.000001 monetary units (sub-paisa), attributable solely to IEEE 754 double-precision float arithmetic order — functionally zero. The platform rounds display values to whole monetary units using Intl.NumberFormat with maximumFractionDigits: 0, eliminating any perceptible error at the interface level.

Goal-split fractions (e.g., Child Education = 0.300752 for Balanced mode) matched the Excel Presets sheet to six decimal places in all cases. This confirms the correct verbatim extraction of irrational fractions that cannot be represented as simple percentages (e.g., 0.300752 ≈ 19/63.2, reflecting the original spreadsheet author's goal-weighting model).

7. User Experience Design

7.1 Design Philosophy

EverBond Wealth's visual language draws from three design lineages: Apple's spatial clarity and generous whitespace, Stripe's fintech data presentation (clean typography hierarchy, soft separators), and Notion's informational minimalism. The palette centres on a warm off-white background (#FAFAF8), gold accents (#C9A84C) evoking premium financial instruments, and four semantic colour channels — sky blue for essentials, rose for risk/emergency, emerald for positive outcomes, and violet for high-risk assets.

7.2 Typography

The platform uses DM Serif Display (Google Fonts) for headings, monetary values, and chart labels — a typeface chosen for its editorial warmth and legibility at large sizes. DM Sans serves as the body and interface typeface, providing optical contrast against the serif display type. All font weights are loaded from a single CDN request using Google Fonts variable font axes.

7.3 Couple Personalization

Partner names are stored in localStorage and surfaced throughout the interface: in the time-aware greeting ('Good Morning, Arup & Shatarupa □'), in the sidebar partnership chip, and as labels for individual salary contribution fields. The greeting logic determines 'Morning', 'Afternoon', 'Evening', or 'Welcome Back' based on the user's local hour — a simple behavioural touch that meaningfully personalises the experience for returning users.

7.4 Feature Comparison

Feature	EverBond Wealth	Generic Budgeting App	Spreadsheet
Couple-native joint income	✓	✗	Manual
Excel-calibrated presets	✓	✗	✓ (static)
SIP compound projection	✓	Partial	✓ (static)
Multi-asset breakdown	✓	✗	Manual
Milestone Planner	✓	Rare	Manual
Live exchange rates	✓	Rare	✗
Mobile Apple-nav UX	✓	Varies	✗

Offline-capable / no login	✓	✗	✓
8-currency support	✓	Partial	Manual
Financial health scoring	✓	Rare	Manual

Table 4: Feature Comparison — EverBond Wealth vs. Generic Budgeting Apps vs. Spreadsheets

8. Limitations and Future Work

8.1 Current Limitations

- Expected return rates in the simulation engine are static presets based on Indian market historical averages and are not dynamically fetched from live market data sources.
- The Milestone Planner uses linear monthly saving without interest accrual; a future version could offer a recurring deposit equivalent with interest modelling.
- The platform is currently optimised for individual couples; expansion to family budgeting with dependant tracking would require significant model extension.
- No tax optimisation layer exists; Indian-specific instruments like ELSS, NPS, and PPF are referenced in allocation labels but their tax treatment is not modelled.
- Accessibility (ARIA labels, keyboard navigation) has not been formally audited; this is a priority for a v3.0 release.

8.2 Future Directions

- AI-driven preset recommendation: Use household spending pattern data (opt-in, device-local) to recommend mode switches or custom allocation adjustments.
- EMI integration: Model home loan and vehicle loan EMIs as fixed outflows within the Essentials envelope, with payoff timeline visualisation.
- Insurance gap analysis: Given income and dependants, compute minimum term life and health insurance coverage requirements.
- Multi-country preset sheets: Commission verified Excel equivalents for US (401k/IRA focus), UK (ISA/pension focus), and Germany (ETF Sparplan focus) to make the platform globally authoritative.
- Progressive Web App push notifications: Monthly reminder to review allocations; anniversary milestone progress updates sent as web push — all computed device-locally.

9. Conclusion

EverBond Wealth demonstrates that a spreadsheet-calibrated, couple-native, browser-native progressive web application can bridge the gap between the mathematical rigour of expert-designed Excel financial models and the accessibility expectations of modern mobile users. By treating the Excel workbook as the definitive source of truth and the web application as its live, personalised, animated interface, the platform achieves a form of financial transparency rarely found in consumer tools: every number on screen is traceable to a specific row and column in the original spreadsheet.

The Milestone Planner extends this philosophy to the shorter-horizon, date-anchored saving goals that characterise real couple financial lives — cars, travel, property deposits — modelling them as complement to, not competition with, the long-horizon investment engine. The integration of milestone contributions into the core budget model represents, to the author's knowledge, the first published implementation of this architecture in open-source personal fintech.

The platform is released as open-source software under the MIT Licence. The author invites the academic community to extend the financial model, add country-specific preset sheets, and study the behavioural impacts of transparent joint financial visualisation on couple financial outcomes — a research agenda with significant real-world welfare implications.

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